

Worked Example — Intro-to-ML (Week 1)

This is the **worked example** that pairs with the how-to guide. It shows the templates *filled in* for one real course (Intro-to-ML Programming). The planning artifacts are complete, and the Week 1 **notebooks are the real ones** — lifted from the course as actually taught. The remaining prose files are stubs that mark where content goes.

The point: someone reading the guide who gets stuck on “what does a *good* X look like?” flips here and copies the pattern.

What’s here

File / folder	What it shows	Depth
01_Decisions.md	The one-page Decisions document, filled in	Complete (it’s short by design)
02_Objectives.md	Course objectives + Week 1 module objectives	Complete
03_Sample_MDD_Week1.md	One week mapped in the MDD (the course-header block + one module table). Timing is deliberately <i>not</i> here — it lives in the teaching guide	Complete for 1 week; rest is “...”
05_Bounding_The_Answer_Space.md	A worked §6a step 5b: how to build a feedback envelope and bank for an assignment with many valid answers	Complete
Week1_folder_skeleton/	The three-package folder structure	Mixed — see the two rows below
.../Student_Materials/day1_The_following_notebook	The following notebook: load → split → fit → score → residual plot, 26 cells, outputs saved	Real (runs end to end)
.../day1_live_session/pair_programming	The same pipeline with eight steps cut out as # TODOs — scaffolding by subtraction	Real

File / folder	What it shows	Depth
.../day1_live_session/images/	The residual plots and the best-fit comparison the guide calls “visual math”	Real
...the remaining 8 prose files	One-line purpose each, marking where content goes	Stubs
Week1_folder_skeleton/Teaching_guide/Day1/Segment1_Fully_Shown/	One-line purpose each, at full depth	Complete (sets the quality bar)

How to read it

1. Start with `01_Decisions.md` → `02_Objectives.md` → `03_Sample_MDD_Week1.md` (the planning artifacts). Then `05_Bounding_The_Answer_Space.md` when you reach §6a step 5b and your assignment has no single right answer.
2. Open `Week1_folder_skeleton/` to see where every file lives.
3. Open **`live_notebook.ipynb`** and **`pair_programming.ipynb`** side by side. This is the single most useful thing here: the practice file is the working file with holes cut in it, which is why it can’t be subtly wrong. You are looking at *build complete, then degrade* in two files.
4. Open `Segment1_FULLY_SHOWN.md` to see the difference between a stub and a finished artifact.

Scope note (on purpose)

The eight remaining prose files are intentionally **stubs**. The notebooks are real because the method’s central move — *build the complete version, then degrade it into practice* — cannot be shown with placeholders; you have to be able to open both files and see the subtraction. Everything else is a stub because filling it would quietly rebuild a whole course, and the structure plus a few real artifacts is enough to communicate “done.”